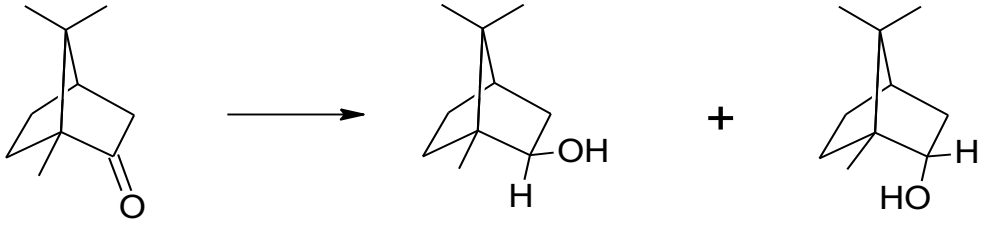
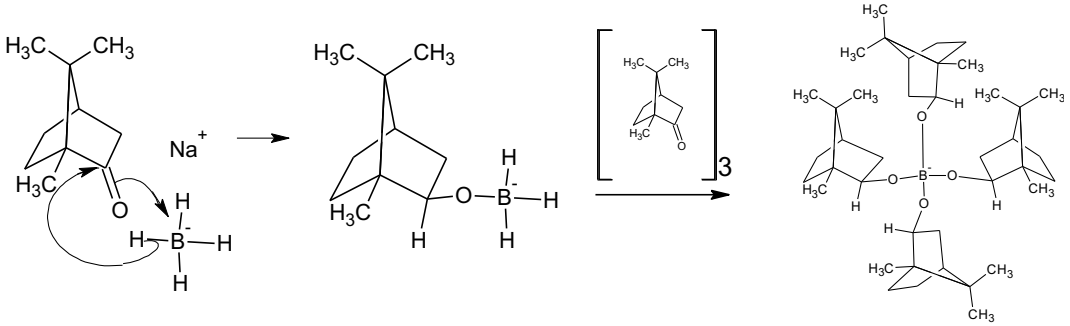
Camphor, a terpene, is found in the *Camphora officinarum* tree which is native to south China and Asia, the *C. camphora tree* ,however, was a primary source for camphor and timber. The compound was used through the centuries in various ways, such as, insect repellent, a food additive, a perfume additive, medicinally in Chinese medicine and as a decongestant, (a familiar ingredient in Vicks Vaporub). Reduction of the carbonyl group of camphor yields two diastereomeric alcohols, isoborneol and borneol, one being the exo form of the alcohol and the other the endo form. Both compounds have similar characteristics and uses.



Camphor Isoborneol Borneol

The reduction of camphor is performed with sodium borohydride, $NaBH_4$. Other metal hydrides such as lithium aluminum hydride, (LAH), can be used; however, LAH is limited to non protic solvents and used under a strict absence of water due to its increased reactivity making its use more complicated.

Due to the steric hindrance from the two geminal methyl groups of camphor, the hydride reaction occurs primarily from the endo approach resulting in the exo isoborneol as the major product. One mole of sodium borohydride has the potential of reducing four moles of camphor molecules.



Camphor

Isoborneol

Week 3.

Reduction of Camphor with Sodium Borohydride (Micro Scale)

1. Weigh out 0.100 g camphor. Transfer it to a 10-mL Erlenmeyer flask.
2. Go to the hood. Add 0.5 mL of methanol to the camphor in the 10-mL Erlenmeyer flask.
3. Stir with glass rod until the camphor has completely dissolved.
4. Keep an ice bath ready.
5. Take about 0.10 g of sodium borohydride in a vial.
6. Cautiously and intermittently add 0.10 g of sodium borohydride to the solution. As the reaction proceeds, the mixture will evolve gas and get warm. When the reaction flask warms to slightly above room temperature, return it to the ice bath.
7. Heat the contents of flask over a warmed sand bath to for 10 minutes.
8. Allow the reaction to cool to room temperature .
9. Carefully add 3.5 mL of ice cold water.
10. Collect the white solid by filtering on a Hirsch funnel.
11. Transfer the solid to a vial.
12. Add 4.5 mL of ether to dissolve the product and add anhydrous magnesium sulfate to dry the solution.
13. Remove the drying agent by putting solution into another weighed vial.
14. Evaporate the solvent

Analysis of Alcohols

1. Determine the weight of the product and calculate the percentage yield.
2. Determine the melting point.
3. Determine the infrared spectrum of the product.

Special Instructions

All the reactants and products are highly volatile and must be stored in tightly sealed containers. Reaction should be conducted in a well ventilated environment or under the hood.

Ether is extremely flammable. There should not be any source of ignition or open flame nearby.

Waste Disposal

The aqueous solutions obtained from the extraction should be disposed of in the designated aqueous container. Any residual methanol or ether should be disposed of in the non-halogenated organic waste container.